Particle Tracking Branch

Overview

The BARCODE platform characterises material dynamics at the *field* level without assigning an identity to an individual particle. The Particle Tracking branch adds a complementary *per-particle* capability, producing individual trajectories and their mean squared displacement.

It does so by reusing the Image Binarization (IB) branch machinery but applying it in three dimensions. Every frame is binarized and stacked into a volume with axes (t, y, x); a single 3-D connected-component labelling is then run over that volume. A particle whose binary island in frame i overlaps its own island in frame $i+1$ becomes one connected 3-D object automatically, so each labelled component is a trajectory and no explicit frame-to-frame association step is required.

Conventional trackers separate detection (locate particles in each frame) from linking (associate detections across frames, typically by solving a nearest-neighbour or global assignment problem). This branch folds linking into the labelling step: linking happens implicitly through spatial overlap along the time axis. The consequence is that the method is only valid when islands displace less than roughly their own size per frame. Below that regime a displacement-based (detect-then-link) tracker is the appropriate tool, and the overlap gate exists to flag exactly this condition.

P.1 Binarization Threshold (offset)

Sets the cutoff used to convert grayscale frames to the binary islands that tracking operates on. Identical to the IB branch: a pixel is foreground when its intensity exceeds the frame mean scaled by $(1 + \text{offset})$,

$$b(x, y, i) = \begin{cases} 1, & B(x, y, i) \geq \overline{B(i)}\,(1 + \text{offset}) \\ 0, & \text{otherwise}, \end{cases} \tag{1}$$

where $\overline{B(i)}$ is the mean intensity of frame i. Lowering the offset lowers the threshold, enlarging each island; raising it shrinks islands. Because island size governs the temporal overlap Φ (P.10), the offset is the primary control over whether consecutive-frame islands overlap enough to link. Default: $\text{offset} = 0.1$ (as in the IB branch).

P.2 Binning Ratio (p_b)

An integer spatial downsampling applied before binarization, identical to the IB branch. Each $p_b \times p_b$ block of pixels is averaged into one:

$$\tilde{B}(x', y', i) = \frac{1}{p_b^2} \sum_{x=(x'-1)p_b+1}^{x' p_b} \sum_{y=(y'-1)p_b+1}^{y' p_b} B(x, y, i). \tag{2}$$

Binning reduces data size and noise but coarsens the coordinate grid by a factor of p_b. Consequently the effective pixel size in physical-unit conversion becomes $p_b\, r$, where r is the micron-per-pixel ratio (µm per pixel), and intensity-weighted centroids (P.7) are disabled for $p_b > 1$ because the grayscale frame no longer aligns with the binary grid. Although the IB branch defaults to $p_b = 2$, quantitative trajectory analysis uses $p_b = 1$.

P.3 Minimum Particle Size ($A_{\min}$)

A per-frame area filter applied to the binary islands before stacking. In each frame, every connected island is labelled and its pixel area computed; islands smaller than $A_{\min}$ are removed:

$$b(x, y, i) \leftarrow 0 \quad \text{for all } (x, y) \in R_{j,i} \text{ such that } |R_{j,i}| < A_{\min}, \tag{3}$$

where $R_{j,i}$ is the pixel set of the j-th island in frame i. This removes background-noise specks at the detection stage, before they can be linked into spurious trajectories. It is the tracking-stack analogue of the minimum-island-size operation of the IB branch. Default: $A_{\min} = 0$ (disabled).

P.4 Connectivity Order (κ)

Sets how permissively two foreground voxels in the (t, y, x) volume are considered connected during 3-D labelling. The value $\kappa \in \{1, 2, 3\}$ is passed to `scipy.ndimage.generate_binary_structure(3,` κ`)`, producing the structuring element used by the label operation:

- $\kappa = 1$ — 6-neighbour (face-adjacent). A voxel connects only to the voxel at the *same* (y, x) in the adjacent frame, or to its 4-neighbours within the frame. An island must therefore overlap at the same pixels across frames to link.
- $\kappa = 2$ — 18-neighbour (adds edge-adjacent). Diagonal steps that combine a one-pixel spatial shift with a one-frame advance now count as connected.
- $\kappa = 3$ — 26-neighbour (adds corner-adjacent). Full spatio-temporal diagonals connect, the most permissive setting.

Higher κ tolerates more inter-frame motion before a trajectory breaks, at the cost of more readily merging near-touching particles. Default: $\kappa = 3$.

P.5 Time Gap (g)

Permits a particle to disappear for up to g frames and still be linked as the same trajectory. Before labelling, the volume is subjected to a morphological *closing* along the time axis, a dilation followed by an erosion, using a one dimensional structuring element S_t of length $g+1$ oriented along i:

$$V_g = (V \oplus S_t) \ominus S_t, \qquad S_t = \underbrace{[\,1,\,1,\,\ldots,\,1\,]}_{g+1\ \text{frames}} \text{ along the } i\text{-axis.} \tag{4}$$

Closing connects trajectory segments separated by up to (g) empty frames without extending the trajectory ends. Larger (g) can bridge longer gaps but may incorrectly link different particles. Particle positions are calculated only from the original detected frames, so no positions are added during gaps. Default: (g=0).

P.6 Spatial Dilation (d)

Permits linking across larger inter-frame displacements by growing each island outward in space before labelling. The volume is dilated within each frame by a square structuring element S_{xy} of half-width d (size $2d+1$ in x and y, unit extent in t):

$$V_d = V \oplus S_{xy}, \qquad S_{xy}(0, \delta_y, \delta_x) = 1 \ \text{ for } |\delta_x| \le d,\ |\delta_y| \le d. \tag{5}$$

An island of extent $w \times h$ becomes $(w+2d) \times (h+2d)$, so two islands separated by up to $2d$ pixels between frames now overlap and link. This extends the method to faster motion when particles are sparse, but merges distinct particles whose separation is less than $2d$. Default: $d = 0$ (no dilation).

P.7 Weighted Centroid

Selects how the position of an island is computed once it has been labelled. Two definitions are available. **Binary centroid** (unchecked), the geometric mean of the island's pixel coordinates, all pixels weighted equally:

$$\bar{x}_{k,i} = \frac{1}{|R_{k,i}|} \sum_{(x,y)\in R_{k,i}} x, \qquad \bar{y}_{k,i} = \frac{1}{|R_{k,i}|} \sum_{(x,y)\in R_{k,i}} y, \tag{6}$$

where $R_{k,i}$ is the pixel set of island k in frame i. **Intensity-weighted centroid** (checked), each pixel is weighted by its original grayscale intensity $B(x,y,i)$, so the centroid is pulled toward the bright peak of the particle:

$$\bar{x}_{k,i} = \frac{\sum_{(x,y)\in R_{k,i}} x\, B(x,y,i)}{\sum_{(x,y)\in R_{k,i}} B(x,y,i)}, \qquad \bar{y}_{k,i} = \frac{\sum_{(x,y)\in R_{k,i}} y\, B(x,y,i)}{\sum_{(x,y)\in R_{k,i}} B(x,y,i)}. \tag{7}$$

The weighted form achieves sub-pixel localization and lower positional noise for compact, bright particles, at the cost of requiring the grayscale frame to be aligned with the binary grid. Default: off (binary centroid).

P.8 Minimum Track Length ($\ell_{\min}$)

A filter on completed trajectories. Since the labelling yields one centroid per component per frame, the number of centroids equals the number of frames in which the component appears. A trajectory T_k is retained only if

$$|T_k| \geq \ell_{\min}. \tag{8}$$

Default: $\ell_{\min} = 2$ (a component must persist across at least two frames to count as a trajectory).

P.9 Minimum Component Size ($V_{\min}$)

A voxel-count filter applied to whole 3-D components after the labels have been masked back to true foreground. The total voxel count of each labelled component is computed, and components below the threshold are removed:

$$\text{drop } T_k \quad \text{if} \quad \sum_i |R_{k,i}| < V_{\min}, \tag{9}$$

This removes small spatio-temporal noise before it forms a false track. Default: $V_{\min} = 0$ (disabled).

P.10 Overlap Threshold and Gate Mode (Φ_{thresh})

The feasibility control. The temporal overlap fraction is computed from the (post-filter) binary stack as the mean fraction of foreground in frame i that remains foreground in frame $i+1$:

$$\Phi = \frac{1}{N_\phi} \sum_i \frac{\sum_{x,y} b(x,y,i)\, b(x,y,i+1)}{\sum_{x,y} b(x,y,i)}. \tag{10}$$

$\Phi \to 1$ indicates islands that barely displace between frames (strong self-overlap, ideal for connectivity linking); $\Phi \to 0$ indicates displacements exceeding island size (no self-overlap, unsuitable). The video is classified as a **good** candidate when $\Phi \geq \Phi_{\text{thresh}}$ and **poor** otherwise.

A companion **gate mode** setting determines the operational consequence of this classification:

- `report` — Φ is measured and the good/poor classification is reported, but linking proceeds regardless. Use this mode to survey a dataset and choose Φ_{thresh}.
- `gate` — linking is skipped entirely (no trajectories returned) when $\Phi < \Phi_{\text{thresh}}$, so that the method refuses to produce tracks in a regime where it is not trusted.

P.11 Trajectory Output and Mean Squared Displacement

Trajectories. Each retained component yields a trajectory $T_k = \{(i, \bar{x}_{k,i}, \bar{y}_{k,i})\}$, one centroid per occupied frame, ordered in time. Trajectories are written for downstream analysis and rendered as a line overlay (`Trajectories.png`) over the field of view, with longer tracks drawn more opaquely so persistent particles stand out from short noise fragments.

Mean squared displacement. The MSD is computed with overlapping time lags and drift subtraction. For a lag of n frames, displacements are pooled over all trajectories and all valid start times,

$$\Delta x = \bar{x}(i+n) - \bar{x}(i), \qquad \Delta y = \bar{y}(i+n) - \bar{y}(i), \tag{11}$$

taken over every pair $(i, i+n)$ in which both positions are defined. The per-lag MSD is the drift-subtracted variance of displacement, scaled to physical units:

$$\mathrm{MSD}_x(\tau_n) = \left(\langle \Delta x^2 \rangle - \langle \Delta x \rangle^2\right) s^2, \tag{12}$$

$$\mathrm{MSD}_y(\tau_n) = \left(\langle \Delta y^2 \rangle - \langle \Delta y \rangle^2\right) s^2, \tag{13}$$

$$\mathrm{MSD}_{2D}(\tau_n) = \mathrm{MSD}_x(\tau_n) + \mathrm{MSD}_y(\tau_n), \tag{14}$$

where $\langle \cdot \rangle$ is the mean over all contributing pairs, $\tau_n = n\,\Delta t$ is the physical lag time, Δt is the frame interval, and $s = p_b\, r$ is the physical pixel size (P.2). Subtracting $\langle \Delta x \rangle^2$ removes any uniform collective drift; for purely diffusive motion $\langle \Delta x \rangle \approx 0$ and the expression reduces to the ordinary $\langle \Delta x^2 \rangle$. The curve is evaluated for increasing n until the number of contributing pairs falls below $n_{\min}$ (default $n_{\min} = 10$).

The results are written to `MSD.csv` with columns `t_lag_s`, `mean_dx_um` ($= \langle \Delta x \rangle\, s$, the drift estimate), `MSDx_um2`, `MSDy_um2`, `MSD_2D_um2`, and `n` (the pair count at each lag).